

COMPARISON OF LIPID DROPLET BINDING PROPERTIES OF TWO EXCHANGABLE PERILIPINS, PERILIPIN 3 AND PERILIPIN 4

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ABSTRACT

Lipid droplets (LDs) are organelles dedicated to lipid storage, composed of a neutral lipid core surrounded by a phospholipid monolayer with embedded proteins. Perilipins (PLINs) are highly abundant LD-proteins, with five PLINs present in mammals. In a cell, different PLINs often do not bind to the same population of LDs, demonstrating differences in their LD-binding properties. We aim to identify the parameters governing PLIN LD binding, focusing on PLIN4, a particularly long and repetitive PLIN. Our previous work identified different sensitivities for lipid packing defects on the LD surface for different PLINs, with PLIN4 and PLIN3 displaying a similar behaviour¹. However, PLIN4 lacks the canonical N-terminal PAT domain present in other PLINs and features an extremely long central amphipathic helix region, making its similar behaviour to the more canonical PLIN3 intriguing. We investigated the contribution of the different domains of PLIN4 and PLIN3 to their LD-targeting. To this end, we expressed full-length and truncated forms of both proteins in human cell lines to explore their LD-binding behaviour. While the central amphipathic region of PLIN4 most strongly affects its LD-binding properties, its non-canonical N-terminal domain also contributes to LD binding. Using *in vitro* approaches and molecular dynamics simulations, we also identified differences in the 4-helix bundle forming C-terminal regions of PLIN4 and PLIN3. Our work sheds light on the contribution of different PLIN domains to their LD targeting.

References:

- 1 A. R. Dias Araújo, et al., *JCB*, **Volume 223**, e202403064, (2024), DOI: 10.1083/jcb.202403064